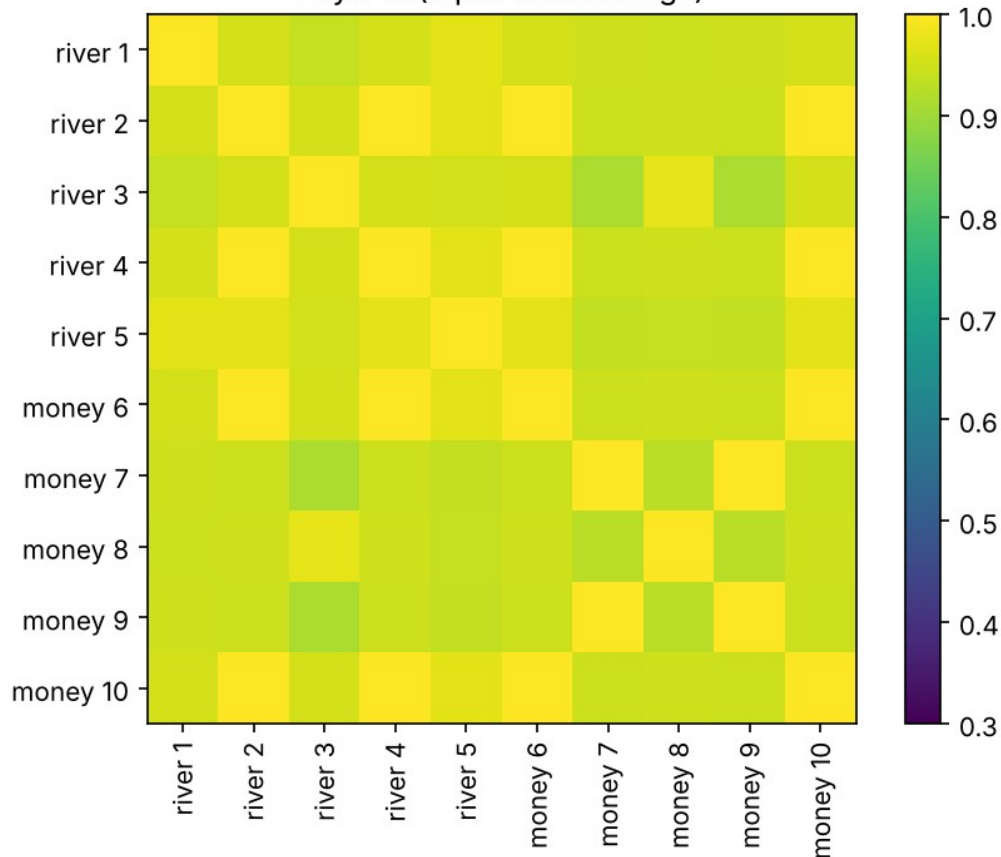
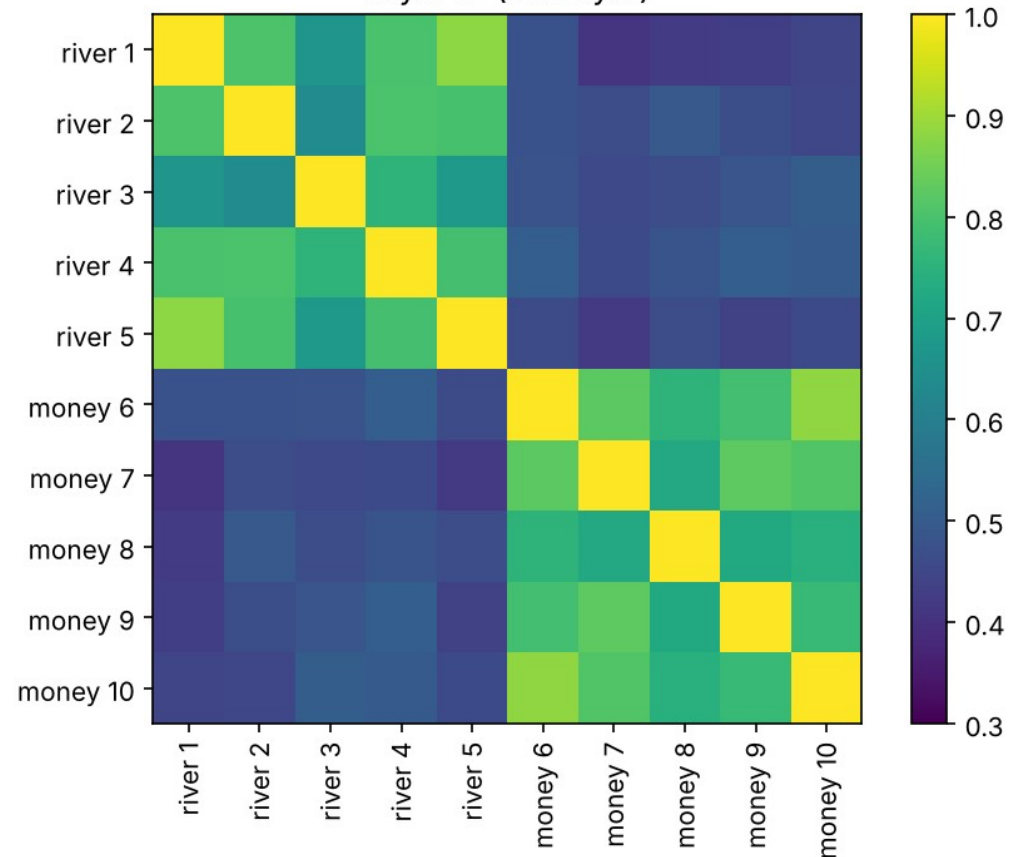


Cosine similarity between occurrences of 'bank'
Two blocks appear on the right: the model has discovered the two senses

Layer 0 (input embeddings)



Layer 12 (last layer)



```
show("ex1_step2_similarity_matrix")
```

```
river = np.array(labels) == "river"
same = (sim[np.ix_(river, river)].sum() - river.sum() + sim[np.ix_(~river, ~river)].sum() - (~river).sum())
same /= (river.sum() * (river.sum() - 1) + (~river).sum() * ((~river).sum() - 1))
diff = sim[np.ix_(river, ~river)].mean()
print(f" mean similarity, SAME sense      : {same:.3f}")
print(f" mean similarity, DIFFERENT sense : {diff:.3f}")
```